

Logistic Regression

From Theory to Practice

A complete theoretical and operational guide: estimating the probability that an input instance belongs to a given category by mapping linear equations through non-linear probability functions.



What Is Logistic Regression?

Definition

A parametric classification algorithm that models categorical output probabilities using input features.

Under the Hood

It computes a raw linear score (z), converts it into Log-Odds, and squeezes it into a probability bounded between 0.0 and 1.0.

$$z = \beta_0 + \sum(\beta_i \cdot X_i)$$

linear combination of weighted features

Input Features

$X_1, X_2, \dots, X_n$ — raw predictor variables

linear combination

Log-Odds (z)

Unbounded score ranging from $-\infty$ to $+\infty$

sigmoid squeeze

Probability ($\hat{y}$)

Calibrated output bounded between 0.0 and 1.0

Every prediction is a probability first — the class label is a threshold decision applied afterward.

Overview of Logistic Regression Types

1

Binary

TARGET OUTCOME

Exactly 2 outcomes (0 or 1)

PROBABILITY ENGINE

Sigmoid Function

EXAMPLE SCENARIO

Disease Diagnosis
(Sick vs. Healthy)

2

Multinomial

TARGET OUTCOME

≥ 3 unordered outcomes

PROBABILITY ENGINE

Softmax Function

EXAMPLE SCENARIO

Customer Segment
(Plan A vs. B vs. C)

3

Ordinal

TARGET OUTCOME

≥ 3 ordered / ranked outcomes

PROBABILITY ENGINE

Cumulative Logit

EXAMPLE SCENARIO

Risk Rating
(Low vs. Med vs. High)

Data Assumptions & Requirements

1

Feature Standardization — Mandatory

Features must be scaled (StandardScaler) so weight coefficients (β) remain comparable and regularization penalties apply uniformly.

2

Linearity of Log-Odds

Assumes a linear relationship between input features (X) and the log-odds of the target (z), not raw probabilities.

3

Multicollinearity Sensitivity

Highly correlated features lead to unstable weights. Mitigated using L2 (Ridge) regularization.

4

Target Encoding

Targets must be numerically mapped depending on model type (Binary $\rightarrow$ 0/1, Ordinal $\rightarrow$ Ranked integers).

Limitations & What to Avoid

1

Non-Linear Data

Fails on non-linear decision boundaries unless polynomial or interaction features are manually engineered.

2

Outliers

Outliers skew log-loss calculations, shifting the decision line away from true patterns.

3

Default Threshold Pitfall

Defaulting to a 0.50 probability threshold leads to poor performance on imbalanced datasets.

4

Severe Class Imbalance

If one class dominates, probabilities shift toward the majority class unless rebalanced (`class_weight='balanced'`).

Model Evaluation Metrics

CONFUSION MATRIX

	Actual Positive	Actual Negative
Predicted Positive	True Positive (TP)	False Positive (FP)
Predicted Negative	False Negative (FN)	True Negative (TN)

Accuracy reads well only on balanced data. On skewed classes, Precision, Recall, F1, and ROC-AUC tell the real story.

Accuracy

Reliable only on balanced datasets.

$(TP + TN) / Total$

Precision

Minimizes False Positives — crucial when false alarms are expensive.

$TP / (TP + FP)$

Recall (Sensitivity)

Minimizes False Negatives — critical in medical or fraud detection.

$TP / (TP + FN)$

F1-Score

Harmonic mean balancing Precision and Recall.

$2 \cdot (P \cdot R) / (P + R)$

ROC-AUC

Evaluates class separability across all probability thresholds.

area under curve

Model Settings: C and Regularization Penalties

$$\text{Total Loss} = \text{Log-Loss Error} + \text{Penalty Term}$$

L1 — Lasso

$$\sum |\beta_i|$$

Shrinks useless feature weights strictly to 0.0. Acts as automatic feature selection.

L2 — Ridge

$$\sum \beta_i^2$$

Shrinks weights smoothly toward 0.0. Keeps all features; handles multicollinearity.

ElasticNet

$$\text{L1} + \text{L2}$$

Hybrid combination blending both penalties for balanced control.

REGULARIZATION STRENGTH — $C = 1 / \text{Regularization Strength}$

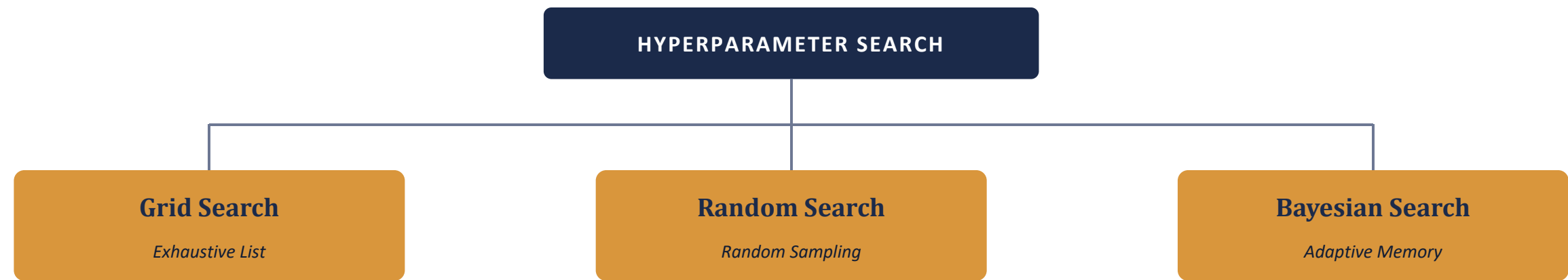
Small C (e.g. 0.01)

Large C (e.g. 100.0)

Heavy penalty → simple model
(prevents overfitting, risks underfitting)

Light penalty → complex model
(fits training data tightly, risks overfitting)

Search Tools: Finding the Winning Configuration



Method	How It Operates	Best Use Case
Grid Search	Tests every combination on a fixed grid — slow on large spaces.	Small datasets, few parameters
Random Search	Randomly samples a fixed number of combinations (n_iter).	Wide log-scale ranges for C
Bayesian Search	Uses past scores to predict high-potential parameter zones.	Complex setups (e.g. ElasticNet) on big data

Log-Odds, Sigmoid, and Log-Loss

1

Linear Log-Odds (z)

z ranges from $-\infty$ to $+\infty$

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$



2

Sigmoid Squeeze

Transforms raw log-odds z into probability $\hat{y}$ from 0.0 to 1.0

$$\sigma(z) = \hat{y} = 1 / (1 + e^{-z})$$



3

Measure Error (Log-Loss)

Heavily penalizes predictions that are confident and wrong

$$\text{Loss} = -[y \cdot \ln(\hat{y}) + (1-y) \cdot \ln(1-\hat{y})]$$



4

Weight Adjustment (Gradient Descent)

Iteratively nudges weights toward lower loss

$$\beta_{\text{new}} = \beta_{\text{old}} - \alpha \cdot \partial \text{Loss} / \partial \beta$$

Binary Logistic Regression: Two-Class Decision Modeling

Target Variable

$$y \in \{0, 1\}$$

Decision Function

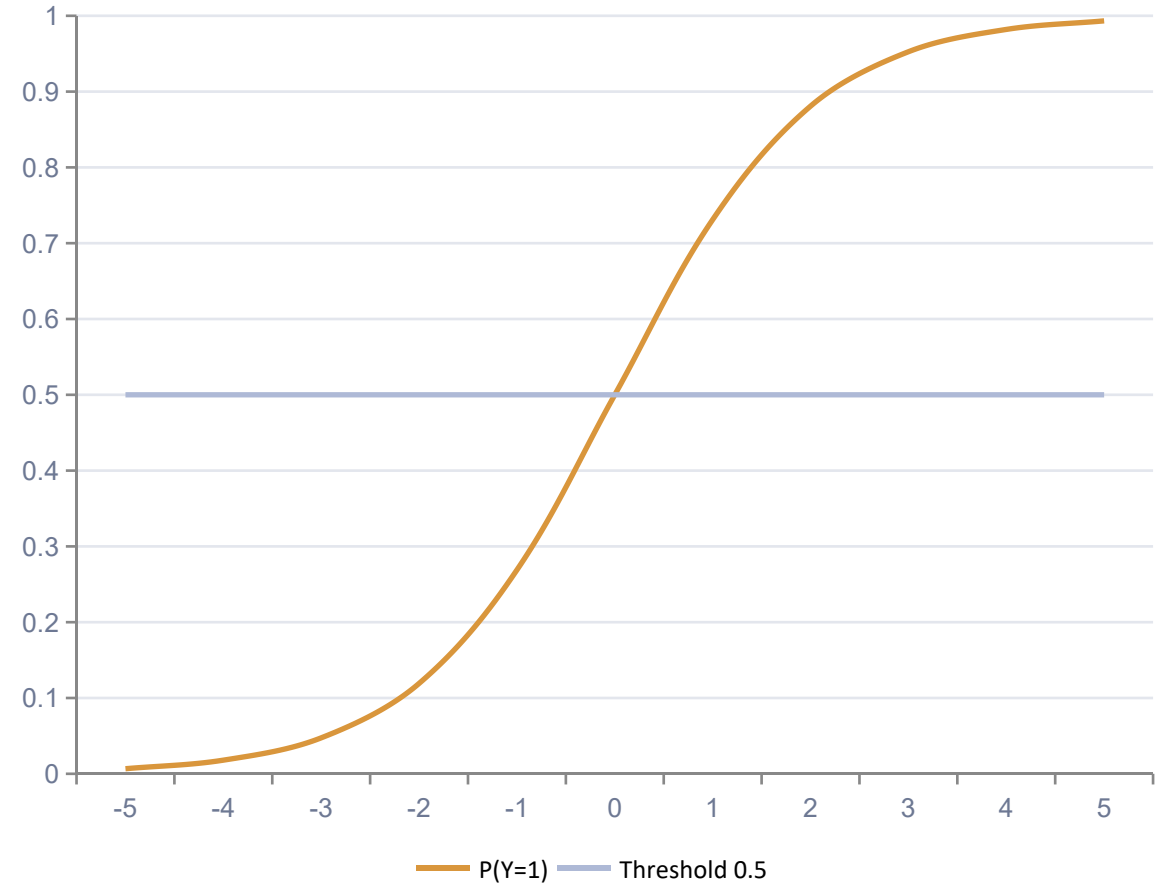
$$\hat{y} = 1 \text{ if } P(Y=1|X) \geq \text{Threshold}$$

$$\hat{y} = 0 \text{ if } P(Y=1|X) < \text{Threshold}$$

How Binary Classification Works

The sigmoid curve maps any log-odds value z to a smooth probability between 0 and 1. At the default threshold of 0.50, the model splits predictions right at the curve's midpoint — but that cutoff can be moved to favor Recall or Precision depending on the cost of errors.

Sigmoid Curve



Banking Credit Card Fraud Detection

PROBLEM STATEMENT

Predict whether an incoming online credit card transaction is Legitimate (0) or Fraudulent (1). Inputs: transaction amount, distance from home location, time since last transaction, device ID risk score.

0.1%

of transactions are fraudulent — a severely imbalanced dataset

DEFAULT THRESHOLD = 0.50

Misses subtle fraud cases (high False Negatives) — costly for the bank.



LOWERED THRESHOLD = 0.15

Catches almost all fraud, maximizing Recall — safer for financial risk.

Why Binary Logistic Regression Works Here

- Computes a continuous fraud probability score (0.0 to 1.0), not just a hard label.
- Allows lowering the threshold from 0.50 to 0.15 to maximize Recall before money leaves the bank.
- L1 (Lasso) penalty removes noisy feature columns automatically.

Binary Logistic Regression: Your Breakdown

YOUR BREAKDOWN

Inputs get separate weights → Calculates z (weighted sum) → Sigmoid squashes z into a probability (0.0 to 1.0) → Operates on linearly separable data → Benefits from standardization → Threshold selection (0.50 default vs. lower/higher) is the key operational limitation



Verdict: 100% Spot On!

Every link in that chain is correct — weights, log-odds, sigmoid, linearity, scaling, and the threshold trade-off are exactly how binary logistic regression operates end to end.

Multinomial Logistic Regression: Multi-Class Unordered Outcomes

Target Variable

≥ 3 unordered classes (e.g. Category A, B, C)

Probability Engine — Softmax Function

$$P(Y=k|X) = e^{z_k} / \sum e^{z_j}$$

Computes probabilities across all K classes so they sum strictly to 1.0 (100%).

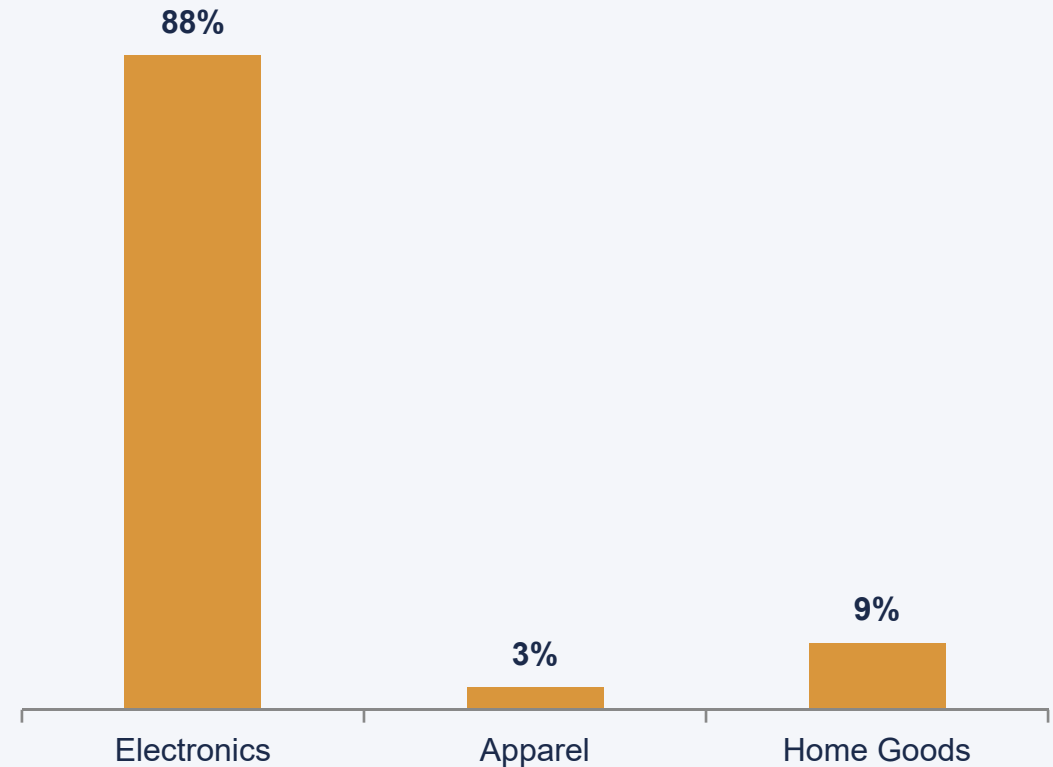
Solvers

Must use multi-class loss solvers (solver='saga' or 'lbfgs').

Imbalance Vulnerability

Highly sensitive to class imbalance — the majority class pulls probabilities toward itself.

Softmax output preview — full case study on the next slide

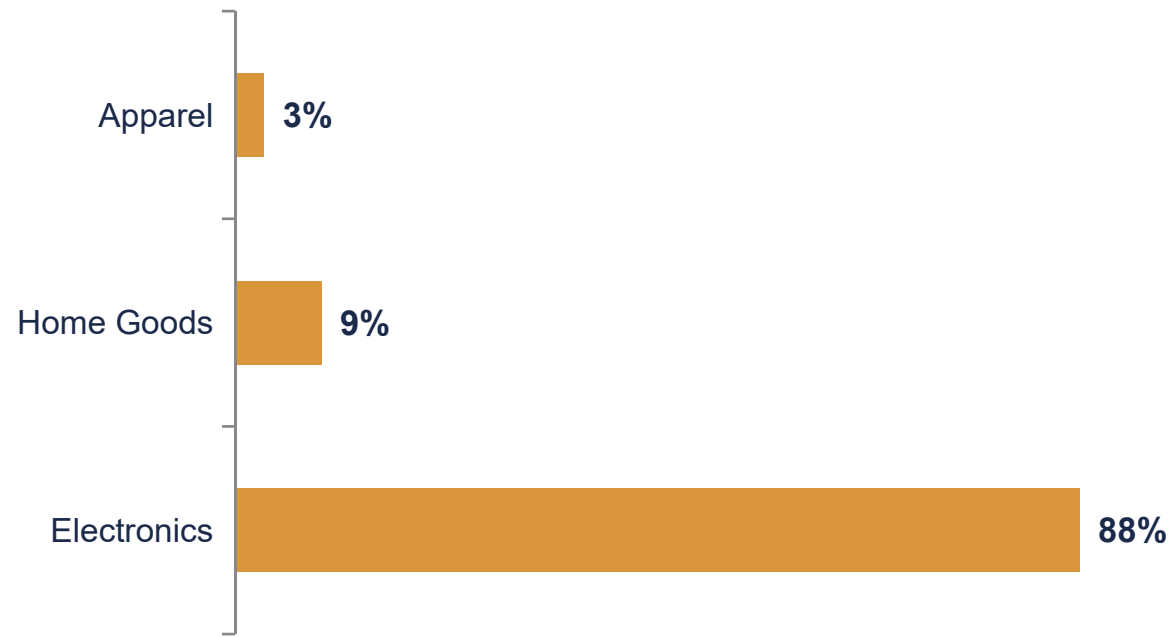


E-Commerce Product Category Routing

PROBLEM STATEMENT

Automatically categorize incoming user search queries into product departments: Electronics (1), Apparel (2), or Home Goods (3). Inputs: search-term word vectors, browsing history, price filter, time on category pages.

Query: "Wireless Noise-Canceling Headphones"



WINNER

Electronics — 88%

Softmax routes the query to the single highest-scoring department.

Sum of probabilities = $0.88 + 0.03 + 0.09 = 1.00$ (100%)

Why Multinomial Works Here

Departments have no numerical ranking — Electronics isn't "greater than" Apparel. Softmax gives a clean probability distribution across all departments simultaneously.

Multinomial Logistic Regression: Your Breakdown

YOUR BREAKDOWN

Handles multiple unordered classes → Uses Softmax (or One-vs-Rest) to compute probabilities for every class simultaneously →
Requires balanced data across classes to avoid bias → Tuned using specific solvers ('saga', 'lbfgs') alongside C and penalty



Verdict: 100% Spot On!

Softmax is indeed just the multi-class generalization of Sigmoid — it ensures the probabilities of all target classes sum strictly to 1.0 (100%).

Ordinal Logistic Regression: Multi-Class Ordered Outcomes

Target Variable

≥ 3 ranked categories (e.g. Low → Medium → High)

Cumulative Probability Engine

$$\text{logit}(P(Y \leq k)) = \alpha_k - (\beta_1 X_1 + \beta_2 X_2 + \dots)$$

Models the probability that an instance falls at or below a given class rank (k).

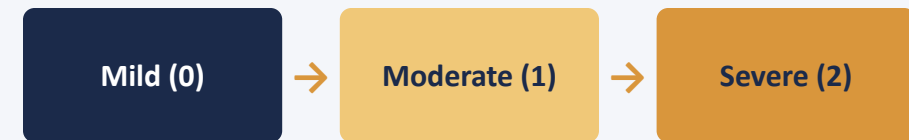
Target Encoding

Labels must be integer-ranked (0 = Low, 1 = Medium, 2 = High). Never use One-Hot Encoding.

Proportional Odds Assumption

Feature weights (β) exert an identical effect across all threshold cutpoints (α_k).

Ordinal Scale: Severity Progression



Split 1 (Cutpoint α_1)

$P(\text{Severity} \leq \text{Mild})$ vs. $P(\text{Severity} > \text{Mild})$

Split 2 (Cutpoint α_2)

$P(\text{Severity} \leq \text{Moderate})$ vs. $P(\text{Severity} > \text{Moderate})$

Preserves the natural medical progression: Mild < Moderate < Severe — the model never treats the classes as interchangeable labels.

Healthcare Disease Severity Grading

PROBLEM STATEMENT

Grade patient heart disease severity as Mild (0), Moderate (1), or Severe (2). Inputs: systolic blood pressure, resting heart rate, cholesterol levels, age, exercise capacity.

SEVERITY SCALE



Why Ordinal Logistic Regression Works Here

- Respects natural hierarchy — predicting Mild when a patient is Severe is a much worse error than predicting Moderate.
- Uses cumulative logit thresholding to keep class ranking intact across every prediction.

Ordinal Logistic Regression: Your Breakdown

YOUR BREAKDOWN

Handles ordered classes → Uses cumulative probability to preserve ranking → Target must be label-encoded to preserve hierarchy (unlike standard one-hot encoding) → Assumes the effect of each feature remains identical across all boundary splits (Proportional Odds Assumption), which often fails in real-world data



Verdict: 100% Spot On!

That Proportional Odds limitation is the exact reason data scientists often turn to Decision Trees when ordinal data gets complex.

Logistic Regression Architecture Summary

1 Preprocessing

- ✓ Scale numeric columns with StandardScaler
- ✓ Encode target: Binary (0/1), One-Hot (Multinomial), or Ranked Integer (Ordinal)

2 Hyperparameter Tuning

- ✓ Select penalty: L1 (feature selection) vs. L2 (multicollinearity)
- ✓ Test C on a log-scale: [0.001, 0.01, 0.1, 1.0, 10.0, 100.0]
- ✓ Choose GridSearchCV (small grids) or RandomizedSearchCV (wide spaces)

3 Model Evaluation

- ✓ Pick metric by business goal — Recall for disease/fraud, Precision for spam
- ✓ Move the decision threshold away from 0.50 for imbalanced data